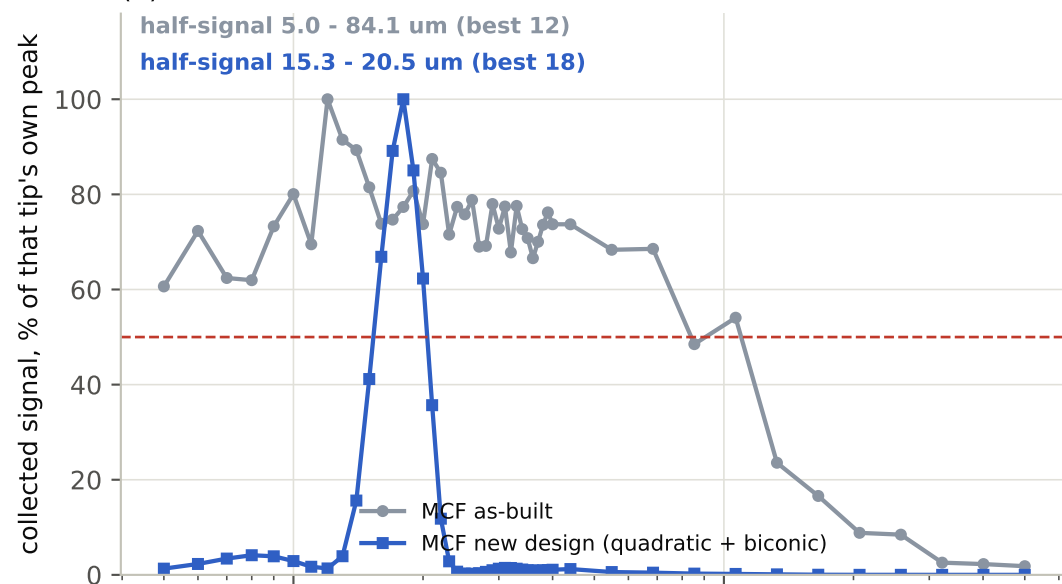
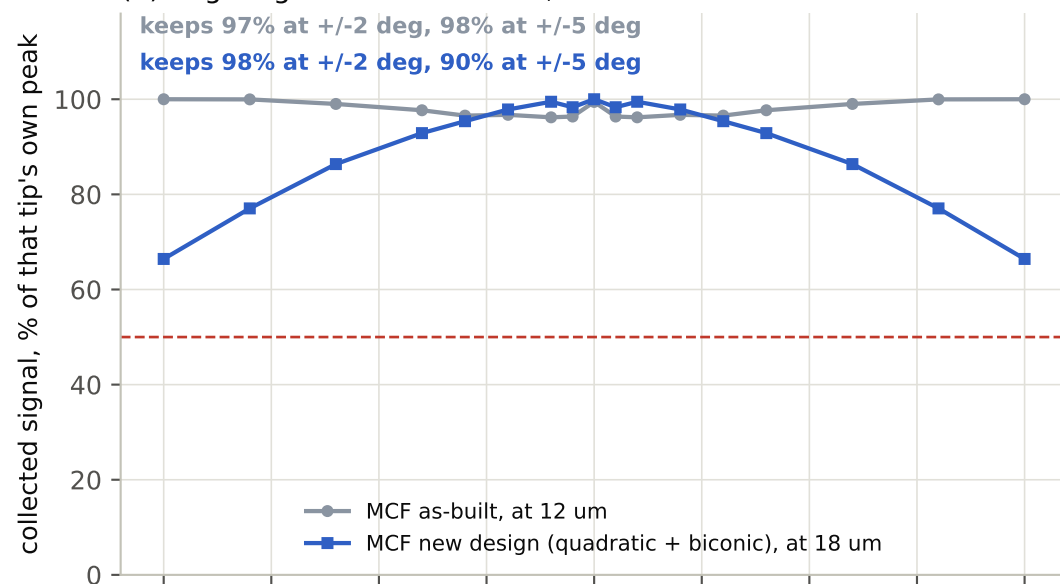


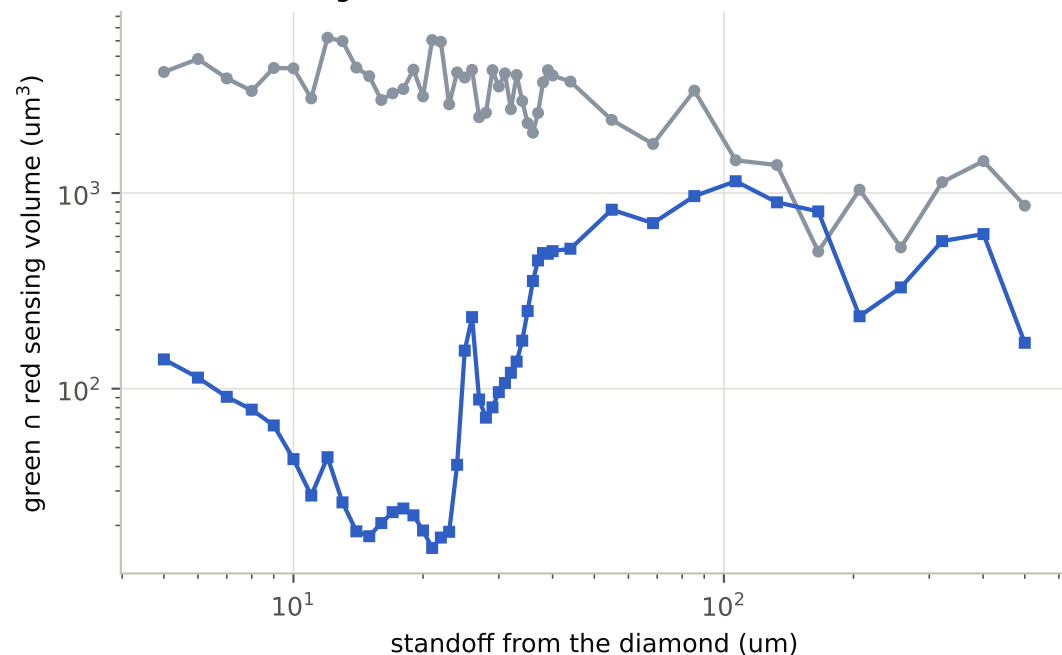
(a) distance from the diamond



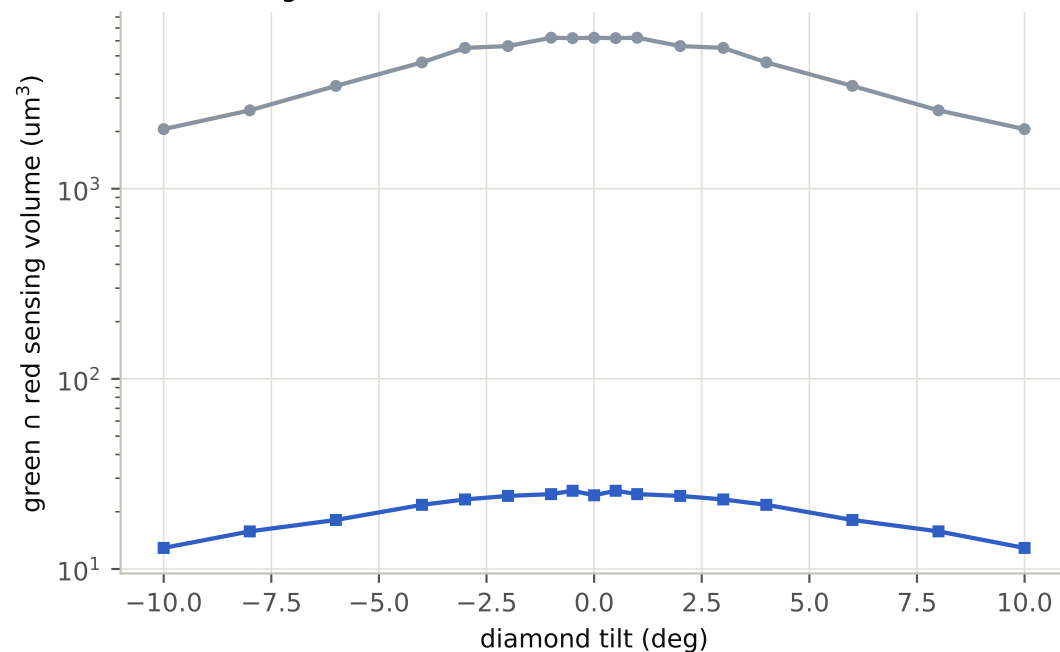
(b) angle against the diamond, each at its own best standoff



(c) volume the green and the red share



(d) same, against tilt



Signal is the product of the traced 532 nm excitation and the six-core collection at every point of the 80-90 um layer, so an NV counts only where the green reaches it and a core can see it; both are re-traced at every standoff and every tilt. (a,b) are normalised to each tip's own peak and compare how fast a tip loses light, not how much it collects -- at its own optimum the new design collects about 195x the as-built tip. Dashed line is half signal. (c,d) give the participation-ratio volume of that product, threshold-free so it stays smooth off-optimum; growing means a blurrier probe, not a better one. Tilt rotates the diamond plane about the fiber y-axis, with all six side cores traced separately. The few-percent ripple on the as-built curve reproduces under grid refinement, so it is the model's ray optics -- hard apertures and no wave averaging -- rather than numerical noise.